

```
# Student Study Hours Analysis

# Create the dataset
students <- data.frame(
  Student = c("A", "B", "C", "D", "E"),
  Study_Hours = c(2, 3, 4, 5, 6),
  Exam_Score = c(58, 65, 72, 81, 88)
)

# View the dataset
print(students)

# Calculate the average study hours
average_study_hours <- mean(students$Study_Hours)
print(average_study_hours)

# Calculate the average exam score
average_exam_score <- mean(students$Exam_Score)
print(average_exam_score)

# Calculate the correlation between study hours and exam scores
correlation <- cor(students$Study_Hours, students$Exam_Score)
print(correlation)

# Create a scatter plot
plot(
  students$Study_Hours,
  students$Exam_Score,
  main = "Study Hours and Exam Scores",
  xlab = "Study Hours",
  ylab = "Exam Score"
)
```